Aggregating Search Logs and Self-esteem for Identifying Individuals with History of Suicide Ideation

Anonymous List of Authors

No Institute Given

Abstract. We propose a framework for identifying individuals with a history of suicide ideation from their daily online search logs. A total of 1,043,855 searches were collected from 120 college students, with an average of 10926 searches per participant, spanning over 4.21 years of search history. The participants also filled out a gold-standard mental health assessment survey. To detect the history of suicide ideation, we first examined both semantic search content category features and temporal recurrence and frequency features. Using those features, we develop a novel encoding that projects these aspects of search behaviors into a low dimensional vector space. Our supervised model incorporates both in-the-moment online behaviors and the ground truth self-esteem for identifying individuals with a history of suicide ideation and verifies the hypothesis that including self-esteem, during modeling, improves the suicide ideation detection task. Labels for the training and testing sets were drawn from professional mental health surveys. Our highest-performing graphical model achieved a classification F1 score of 0.83 with an AUC of 0.81 on stratified test set population. Our framework of leveraging passively sensed search data could help caregivers and practitioners to identify individuals with a potential history of suicide ideation and extend help.

Keywords: history of prior suicide ideation · low self-esteem· mental health · youths · college students · individual search logs · online search history

1 Introduction

It is estimated that someone dies by suicide somewhere in the world every 40 seconds[1], making suicide a leading cause of death worldwide [12]. Globally, suicide ideation (SI) and non-fatal suicide attempts have an approximate life time prevalence of 9.2%, compared to 2.7% for suicide attempts [26]. SI is a strong predictor of suicide attempts and deaths, resulting in a series of negative consequences such as injury, social stigma, hospitalization, and financial burden, *etc.* [26,27,44].

[1] `https://bit.ly/2QSuJkf`

An important challenge in suicide prevention efforts is to identify individuals most at risk so that appropriate interventions and help can be provided early enough. Traditionally, healthcare providers only learn that a person is at risk of suicide when that person comes into contact with medical or mental health services or reaches out to a suicide hot line or others. However, this traditional method fails to cover much of the population at risk. Most obviously, it does not reach people who infrequently employ medical care. This is particularly problematic for teenagers and young adults, who are simultaneously the most physically healthy demographic, consequently seeking care infrequently, yet one that is at significant risk of suicide.

In this work, we leverage online behavior logs that is ubiquitous for all internet users, and is less likely to be subject to self-censorship: web searches. The average internet user uses Google search at least once a day, and many search dozens of times a day [14]. Several researchers have shown that search engine logs can be used as proxy for assessing population level suicide rates [19,1]. One of the biggest shortcomings of these studies is that they only focused on population-level search logs and failed to identify vulnerable individuals. Secondly, they do not connect mental health phenomena such as self-esteem and history of suicide ideation with search engine logs.

Search histories can provide a window into a user's thoughts, concerns, interests, and emotions at a daily, or even hourly, level of granularity. We hypothesize that a user's SI and related mental health concerns can be inferred by analyzing this search history. Furthermore, one's search history presents a dynamic picture of the user over time. We hypothesize that the detection of prior suicide ideation (PSI) can be strengthened by incorporating these dynamics into a model. In order to explore these hypotheses, we gathered search logs from consenting subjects along with their answers to a mental health survey, following the privacy-protecting protocols pioneered by our prior work [48]

We note that SI can be a result from the complex interplay among biological, psychological, social, and environmental factors [42]. While studying all of these factors is beyond the scope of this paper, we do consider the important factor of self-esteem. Low self-esteem can trigger SI [10], and its presence can differentiate between mere suicide ideators from attempters [45]. Our models therefore incorporate self-esteem while detecting PSI. To the best of our knowledge, we are the first to empirically demonstrate the interplay of ground truth SE and ubiquitous daily search logs in detecting individuals with history of SI. As we shall see, the best model achieves an F-score of 0.83 for detecting PSI with the help of SE knowledge.

The main contributions in this paper are:(i) the first demonstration that prior suicidal ideations can be detected from search logs, the most common kind of user internet activity [33]; (ii) new methods for modeling the temporal aspects of search logs; (iii) a novel model-agnostic vector representation of users' online activities; (iv) an innovative graphical model that fills in the gap of connecting in-the-moment online behavior and self-esteem to infer the history of suicide

ideation; (iv) computational verification of the hypothesis that self-esteem can be a correlated factor of suicide ideation.

2 Background

Relationship between self-esteem and different mental health conditions such as depression [39,29], anxiety [41,38], loneliness [43], self-harm, and suicidal thinking has been extensively studied in the research community. For example, the association between self-esteem and suicidal ideation [8,25], how suicidal ideation is significantly related to level of self-esteem [9], and how low self-esteem and depression are independently associated to suicide ideation and attempts [45] are just to name some of the highlighted works in this area. However, none of these studies leveraged online activities log from search engines as the data source of their study.

There is much prior work on SI detection using social media data [7,6,16,23]. [13] and [18] provides a comprehensive summary of SI detection using machine learning approaches on social media data. It is important to note that many of the studies on social media data are limited by the lack of ground-truth data, or are case studies with very small sample size.

With the exception of [48], previous work based on web search data was designed to find trends at a population level rather than to identify individuals at risk of suicide. Researchers used publicly available Google Trends data to assess various mental health conditions such as identifying seasonality in seeking mental health [3], monitoring depression and suicide related terms to help predict rates of self-injury and death within the population [40,46,17]. Nuti *et al.* (2014) provides a through review of the usage of Google Trends in the health care domain [28].

We do note that Internet searches associated to suicide and other related phenomena are not necessarily direct indicators of likely suicide behavior [30]. In addition to the ethical concerns due to sensitive nature of the data, there are significant practical challenges associated to using internet based prevention and interventions. Bailey et al. (2020) provides a comprehensive overview of the issues around employing internet based interventions for suicide prevention [4]. However, our research team, which includes medical professionals active in suicide prevention research, shares the intuition that tools for detecting history of SI would be a valuable public health framework both for mass screening and for use with a population during or post treatment for mental health conditions.

3 Methods

The data collection process for this work has been adapted from [48], see Fig. 1. We acknowledge the proprietary nature of individual search history and employ an Institutional Review Board approved, publicly accessible cloud-based

application for acquiring consent and downloading anonymized, HIPAA compliant search log data using the publicly available Google Takeout interface, as described below, [15].

Google Takeout: Through this platform, Google enable every user to access and download his/her online activities with the click of a few buttons. Using the Google account (i.e, Gmail) associated to an user, Google aggregates and ties all online activities, originating from all different sources (mobile, laptop, tablets, etc.), and make it accessible via the Takeout platform. As long as one is logged into his/her Google account, all engagements on Google products are logged and stored in an unified location at Google's end.

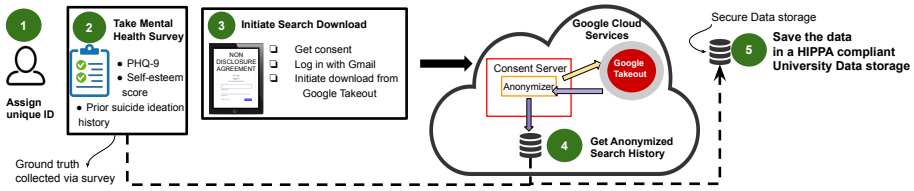


Fig. 1. Individual level Google Search engine activity history can be downloaded using the Google Takeout interface. Prior to any data analysis, the data has been de-identified and anonymized via Google DLP API

3.1 Data Collection

Study population: We recruited participants from our college campus. The data collection process spanned over 14 months, from July 2018 to September 2019 to help reflect seasonal variation typical of academic settings. The study participation was voluntary, and participants received a $20 gift cards in compensation for their time in completing the study. Overall, 160 individuals took part in the study. But only 120 participants resulted in having complete and valid data. Hence, the sample size for this study is 120.

Mental health ground-truth via survey: The ground truth data about an individual's history of suicide ideation and low self-esteem is measured via a clinically validated the mental health survey using the Patient Health Questionnaire (PHQ) [22] and Rosenberg self-esteem (ROS) [35] scales, respectively. The 30-point ROS scale is a commonly used tool to assess levels of self-esteem among young adults [32]. A ROS score of below 15 suggests the prevalence of low self-esteem. The PHQ is commonly used psychological instrument for screening depression [47] and other related mental health phenomena. One of the questions in our PHQ survey asks the question: *Have you ever, in your whole life, thought to kill yourself or consider committing suicide?* Participants answering

yes to the question are labelled as subjects with past suicide ideation. Individuals who answered yes to the history of suicide ideation question were also asked (i) when they had the self-harm ideation; (ii) if they had consulted a therapist or counselor for help. The response from all subjects with past suicide ideation condition showed that ideation occurred within the last 2 years (from the day of our study), a timeline that overlaps with the last years of high school and early years of college. For this reason, we limit the online data used for analysis to be the most recent 2 years of one's online activities. 68% (34 out of 50) of the subjects with history of suicide ideation have admitted that they had gone to a therapist or a counselor to help address their problem. Given our population sample consisted of college students, such high incidence of prior suicide ideation is not surprising because it has been reported that young adults from high school [24] and college [2] are more likely to seriously consider attempting suicide and 22% of this population reported suicide ideation in the past 12 months [20].

Based on the survey response from the ROS questionnaire, 56 individuals were screened positive for low self-esteem and 64 reported no sign of low self-esteem. Following the responses from the PHQ, 50 participants were screened positive with the evidence of the prior suicide ideation (PSI) and 70 were screened negative (NPSI). 3 individuals did not completely finish the PHQ survey; 28 individuals either did not have search data or used Google Search in foreign language; 9 did not have search history longer than 2 months and were excluded from the associated analysis. Table (a) in Fig.2 summarizes the demographic breakdown of the participants in this study. Table (b) & (c) in Fig.2 presents the self-esteem and history of suicide ideation distributions of the subjects based on the ground truth survey responses.

Demographic Variable		Value
	Age	20.93 (mean)
Gender	Male	45.00 % (54)
	Female	55.00 % (66)
Sexual Orientation	Heterosexual	84.17 % (101)
	Bisexual	8.33 % (10)
	Homosexual	4.17 % (5)
	Other	3.33 % (4)

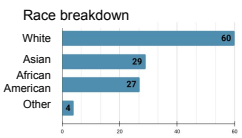
(a)

# Subjects with Prior Suicide Ideation(PSI)	50
# Subjects with No Prior Suicide Ideation (NPSI)	70
Total	120

(b)

# Individuals with Low Self-esteem (LS)	56
# Individuals with no Low Self-esteem (NLS)	64
Total	120

(c)



	# PSI subjects	# NPSI subjects	Total
Training set	39	57	96
Test set	11	13	24
Total (n) for this study			120

(d)

Fig. 2. (a) Participants demographics for this study (N = 120, excluding missing and incomplete data). (b) & (c) Distribution of subjects with/without past suicide ideation and self-esteem condition based on survey responses, respectively. (d) shows the complete data breakdown used in this study.

Online behavior data via Google Takeout: Using the Google Takeout platform, we downloaded Google Search activities history for every participants.

Prior to sharing the data with research team we anonymized and de-identified the search logs, as described below. We only considered search logs from participants who used Google services in English and completed the survey. Overall, 835, 084 searches were collected from all the participants, with an average of 9077 searches per participant, spanning over 4.35 years of search history. As mentioned before, we only use the most recent 2 years of Google Search logs as online data for this study.

Once an individual participates in the study and shares his/her Google search history, the research team receives a one time dump of their anonymized online history. Participants always had the option to opt-out of the research study at any point of time. However, we did not have any such case.

Anonymizing Search Logs: Since much of our life is lived online, people seek the help of popular search engines, such as Google, Bing, *etc.* for virtually almost everything. At times, either intentionally or accidentally, one may search for things that may reveal sensitive information about a user. For example, name, email, phone number, social security, financial information (credit and banking information), address such as physical, internet protocol (IP), media access control (MAC) address, *etc.* We treat search logs as sensitive information and remove any potential personal identifiers prior to making the data available for the research team to analyze. We use Google's state-of-the-art Data Loss Prevention (DLP) API to identify and redact sensitive text from the search histories [31].

Search Log Labelling: Search phrases are typically succinct, recorded in the moment, and the context of the search can be good indicators of what may be going through an individual's mind at an instant of time. We use the the content classification feature of the Google Cloud NLP API [2] to classify each search query. The API returns a hierarchical category label for every search query. We treat the broad category within the hierarchy as the "category" label for the query. For instance, for a query q, if the label from the API is "News/Sports/Soccer", we consider "News" as the category for q. In total, there are 27 such top-level categories, see Fig. 3 (a). In the rest of the paper we will refer to the notations in Fig. 3 (b)

3.2 Data Breakdown

Prior to any analysis, feature engineering, and modeling, we separate the dataset into two parts, namely the **training set** and **test set**. We employ a random stratified data splitting method with training-testing ratio of 80 : 20. The training set consists of 90 (80% of 120) individuals (39 PSI and 57 NPSI). The test set has 24 (20% of 120) participants (11 PSI and 13 NPSI). Table (d) in Fig.2 shows a comprehensive breakdown of the data used in this study.

[2] `https://cloud.google.com/natural-language/docs/classifying-text`

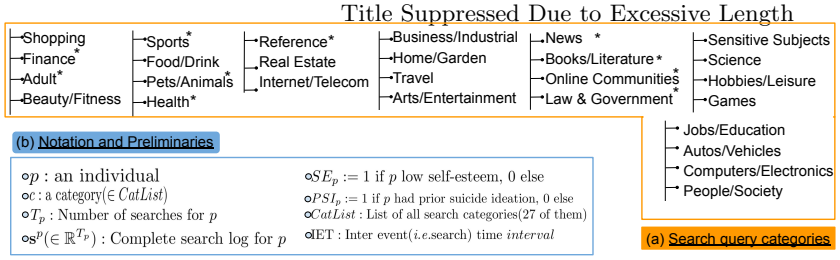


Fig. 3. (a)All search queries are labelled as one of the above 27 broad categories using Google Cloud NLP API. However, only the categories (10 of them) with * are the ones we have identified that are statistically significantly different (with $p < 0.05$) between individuals with/without PSI and NPSI on the training set. For modeling purpose, we use only the categories with * next to them. (b) Notation and Preliminaries for the rest of the paper.

3.3 Identifying Signals from Google Search logs

In this section we define generalizable signals from search logs which can be later used as features for the modeling purpose. We use three types features: Category features, that summarize the semantic meaning of online searches; Lambda features, that summarizes the temporal distribution of categories; and SE-augmented Lambda features, which combine the Lambda features with a binary self-esteem feature derived from the survey data. A pictorial overview of the feature engineering process can be found at this link [3].

Using only the training set, we first eliminate categories that have weak signals in distinguishing between PSI and NPSI. Specifically, we look for categories such that the number of searches labeled with that category is statistically significantly different between the PSI and NPSI groups. Given 96 subjects in the training set, we run a Mann-Whitney U test on each of the categories and identify only those with p value < 0.05, for us we found 10 such categories. Fig.3 lists all the categories, and the 10 with $\star$ next to them are the significant ones. Below we present the three types of features extracted using these 10 significant categories.

Category Features For a person p, we get a 10 dimensional vector representing the distributions of search categories. Each dimension of this vector corresponds to a specific category. Each entry in this vector represents the relative frequency (in percentage) of searches belongs to that category across the whole online history of p. We treat this as the baseline representation ($\in \mathbb{R}^{10}$) of search logs.

Observation: The features obtained using the distribution of search categories was statistically significantly (with p value < 0.05) different between individuals with/without low self-esteem, a result that is consistent with our prior work [48]. However, this representation does not include the temporal aspect of the search logs.

[3] `https://bit.ly/34Ygwt6`

Lambda Features Unlike the Category Features, here we propose a representation that consolidates both the semantics and temporal nature of searches. From Fig. 4(a), we observed that the time intervals between consecutive searches, *i.e.*, inter-event time (IET), appear to exhibit an exponential distribution. This motivated us to consider the IET for each search category, and we found that these IETs tend to follow similar exponential behavior. For a person p and category c, we compute IET(c) by considering the time between every consecutive searches belonging to category c. In layman's term, given the search history of a person, for every category c, a list of inter-event time duration between consecutive searches from c has been computed. Overall it returns a list of lists of inter-event times, where each inner list corresponds to the inter-event times of a particular category.

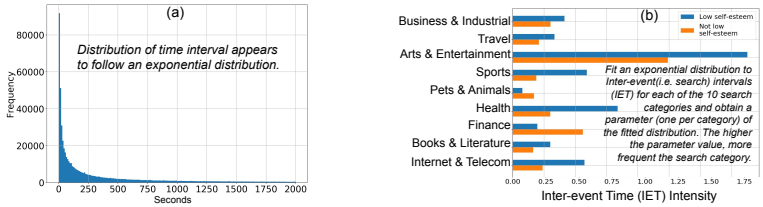


Fig. 4. (a) Distribution of time interval between consecutive search queries.(b)Inter-event(i.e. search) intervals (IET) for each of the 10 search categories for which the parameters seem to be significantly different for the people with/without low self-esteem.

After that, for each $c \in CatList$, we fit an exponential distribution to $IET(c)$ and obtain a parameter, λ_c. An exponential distribution with parameter λ is described by the pdf: $f(x) = \lambda \exp(-\lambda x)$. Since there are 10 categories, we get 10 such parameters. We call this the Lambda Features, $\boldsymbol{\lambda} = (\lambda_1, \ldots, \lambda_{10})$

One way to interpret the Lambda Features for a category is that it encapsulates the frequency of recurring searches from that category, in addition to the semantic context of the query. More precisely, the Lambda Features can be treated as an event rate. The higher the value of lambda, the more recurrent (smaller IET) is the searches from the respective category. In this way, one can consider Lambda feature as a concise spatio-temporal encoding of search logs.

Observation: For most of the search categories, the corresponding lambdas were significantly larger for people with low self-esteem. In other words, when an individual with low self-esteem queries Google, they tend to do more searches related to the most recent query within a short time interval. This is demonstrated by the difference in lambda values between categories such as "Health", "Sports", "Arts & Entertainment", *etc*, see Fig.4(b). It is arguable, but one plausible explanation can be that individuals with low self-esteem are more reliant on search engines to seek reassurance about everyday things such school, office works, *etc*. We think that this may be, at least in part, a reflection of individual search acts as a form of validation. Prior work has shown that individuals with high self-esteem do not need continual validation [21]. It is important to note that our observed shorter IETs in online search activities among low self-esteem

individuals can be a result of other confounding factors such as history of attention deficit hyperactivity disorder (ADHD), especially since it has been reported that between 2% and 8% of college students, a population similar to ours, are estimated to have ADHD [11].

SE-augmented Lambda Features In this representation, we augment the ground truth binary self-esteem (i.e. SE) label with the 10 dimensional Lambda Features from previous section, resulting in a 11 dimensional embedding of search logs. We set the binary SE label to be 1 for a participant with ROS score < 15 (having low self-esteem), otherwise SE is set to 0. This allowed us to include an individual's search behavior and ground truth self-esteem information in a more comprehensive and straightforward way.

To summarize, with the 10 statistically significantly differentiating categories that separates the PSI and NPSI groups, we developed Category Features ($\in \mathbb{R}^{10}$) and Lambda Features ($\in \mathbb{R}^{10}$) from online activity logs. We concatenate the Lambda Features with self-esteem information to obtain SE-augmented Lambda Features ($\in \mathbb{R}^{11}$). It is important to note that, for all modeling proposed in our work, we either use Category Features ($\in \mathbb{R}^{10}$) or Lambda Features ($\in \mathbb{R}^{10}$) or SE-augmented Lambda Features ($\in \mathbb{R}^{11}$). We do not concatenate or mix these features together but only compare between them.

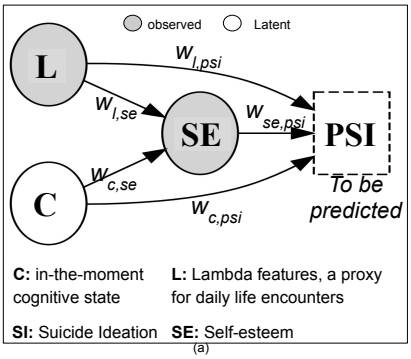
4 Modeling and Results

For the PSI detection task, we ask the question: *"Given the search logs and the ground truth self-esteem label of an individual, can we identify whether the person exhibited a history of suicide ideation?"*

Baseline Classifiers Using the three proposed feature representations from the Feature Engineering section, we train a Logistic Regression and a linear Support Vector Machine (SVM) for the history of suicide ideation detection task. We evaluate the performance of these models using the held-out test set mentioned in Data Breakdown section.

Graphical Model Observe that the baseline models that have been proposed in the previous section do not leverage the latent inter-dependencies between online search behavior and ground truth self-esteem (SE). However, Wild et al. (2004) showed that self-esteem can be a strong predictor of SI [45]. Furthermore, Rosenberg et al. (1986) have proposed that self-esteem, at any given point of time, assumes a baseline cognitive state (C) in absence of external factors [36]. This default mental state is unobserved and gets updated by the daily experiences in life. Accordingly, we designed a graphical model to capture these dependencies, see Fig.5. We treat searching behavior (represented via Lambda Features as L) as a proxy for day-to-day encounters which may give rise to observable mental states such as low self-esteem (SE) and prior suicide ideation

(PSI). Hence, there are direct edges from C and L to SE. Similarly, there are direct edges from C and L to PSI.



(b) Assumptions regarding Model Parameters

$$C \sim \mathcal{N}(0, \mathbf{1}_k) \tag{1}$$

$$SE \sim \mathrm{B}\left(\sigma(\mathbf{w}_{c,se}^T C + \mathbf{w}_{l,se}^T L)\right) \tag{2}$$

$$PSI \sim \mathrm{B}\left(\sigma(\mathbf{w}_{c,psi}^T C + \mathbf{w}_{l,psi}^T L + w_{se,psi} SE)\right), \tag{3}$$

where σ is the *Sigmoid* function, B denotes the *Bernouli* pdf, and $\mathbf{w}_{c,se}$, $\mathbf{w}_{c,psi} \in \mathbb{R}^k$ & $\mathbf{w}_{l,se}$, $\mathbf{w}_{l,psi} \in \mathbb{R}^{10}$ and $w_{se,psi} \in \mathbb{R}$ are the model parameters.

(c) Model Parameter Estimation

$$\begin{aligned} &\mathbb{P}\left[\mathbf{w} | L, SE, PSI\right] \\ &\propto \underbrace{\mathbb{P}[\mathbf{w}]}_{Prior} \cdot \underbrace{\int_C \mathbb{P}[SE, L, PSI | C, \mathbf{w}] \cdot \mathbb{P}[C] dC}_{Likelihood} \\ &= \underbrace{\mathbb{P}[\mathbf{w}]}_{Prior} \cdot \underbrace{\int_C \mathbb{P}[SE | C, L, \mathbf{w}] \cdot \mathbb{P}[PSI | SE, C, L, \mathbf{w}] \cdot \mathbb{P}[C] dC}_{Likelihood} \end{aligned} \tag{4}$$

Fig. 5. (a)Graphical model for the history of SI detection task using search log and ground truth self-esteem label; (b) Assumptions about the model; (c) Parameter Estimation

Assumptions Regarding Model Parameters: Next, we describe the assumptions about the proposed graphical model. We assume $C \in \mathbb{R}^k$ (where $k = 1$ is a hyperparameter achieved after tuning), $L \in \mathbb{R}^{10}$ and $SE, PSI \in \{0, 1\}$. The dependency between the nodes are presented in Fig. 5(b).

Parameter Estimation: We denote $\mathbf{w}$ as the set of all model parameters, where $\mathbf{w} = (\mathbf{w}_{c,se}, \mathbf{w}_{c,psi}, \mathbf{w}_{l,se}, \mathbf{w}_{l,psi}, w_{se,psi})$. The parameters are estimated from the posterior distribution. Using Bayes rule, the posterior distribution for the given graphical model can be computed as shown in Fig. 5 (c):

The prior and likelihood can be computed from (1), (2), and (3) in Fig. 5(b). The posterior distribution is inferred using Markov chain Monte Carlo (MCMC) method. We use the package PyMC3 [37] for this task.

Benefit of Graphical model over SVM and Logistic Regression: We infer the baseline mental state from the training data. From the structure of graphical model we can see that once the baseline state and the model parameters are inferred, the PSI label can be obtained only by using the online search data. Hence, the graphical model, during testing phase, does not require the ground truth SE score, unlike the baseline models.

As previously mentioned, we randomly split the 120 subjects into a training set and a test set with ratio of 80 : 20 before any feature engineering, modeling, or analysis. We ensure the split is stratified with respect to the class percentages of the PSI/NPSI subjects, as shown in Fig. 2 Table (d).

First, we trained two baseline supervised models, linear SVM and Logistic Regression, on the task of PSI detection with Category Features, Lambda Features, and SE-augmented Lambda Features, respectively. We present the performance on the test set of these two baseline models with each of the features in Fig. 6. The detailed precision, recall, and F1 scores for each class and ROC AUC of each baseline classifier are also included.

Test set N = 24	PSI	NPSI	Avg.	95% CI
Precision	<0.61> (0.57)	<0.67> (0.67)	<0.64> (0.62)	<0.60, 0.68> (0.57, 0.66)
Recall	<0.85> (0.92)	<0.36> (0.18)	<0.60> (0.55)	<0.56, 0.65> (0.51, 0.60)
F1 score	<0.71> (0.71)	<0.47> (0.29)	**<0.59>** (0.50)	<0.55, 0.64> (0.45, 0.54)

(a) Category Feature

Test set N = 24	PSI	NPSI	Avg.	95% CI
Precision	<0.75> (0.71)	<0.67> (0.70)	<0.71> (0.71)	<0.67, 0.75> (0.67, 0.75)
Recall	<0.69> (0.77)	<0.73> (0.64)	<0.71> (0.70)	<0.67, 0.75> (0.66, 0.74)
F1 score	<0.72> (0.74)	<0.70> (0.67)	**<0.71>** (0.70)	<0.67, 0.75> (0.66, 0.75)

(b) Lambda Feature

Test set N = 24	PSI	NPSI	Avg.	95% CI
Precision	<0.77> (0.73)	<0.73> (0.78)	<0.75> (0.76)	<0.71, 0.79> (0.72, 0.79)
Recall	<0.76> (0.85)	<0.72> (0.64)	<0.74> (0.74)	<0.70, 0.78> (0.70, 0.78)
F1 score	<0.76> (0.79)	<0.74> (0.70)	**<0.75> (0.74)**	<0.71, 0.79> (0.70, 0.78)

(c) SE-augmented Lambda Feature

	Category Feature	Lambda Feature	SE-augmented Lambda Feature
ROC AUC	<0.60> (0.55)	<0.71> (0.70)	**<0.75> (0.74)**

(d) AUC ROC

Note:
→ Numbers within < > are for Linear SVM
→ Numbers within ( ) are for logistic regression

Fig. 6. Numbers within ⟨ ⟩ are for linear SVM and numbers within ( ) are for Logistic Regression. Table (a) & (b) shows the performance of the SVM and Logistic Regression model using only the Category and Lambda Features extracted from online activities, respectively. Notice the model performance increase to 0.71 in Table (b) from 0.59 in Table (a). Table (c) shows the performance of the models when using self-esteem and Lambda Features only. Table (d) shows the AUC ROC for each of these features. Notice that the AUC is highest when SE-augmented Lambda Features are used.

Category Features are frequency-based whereas Lambda Features incorporates the spatio-temporal aspect of search logs. None of these two has the SE knowledge embedded. The SE-augmented Lambda Features ($\in \mathbb{R}^{11}$) contain a comprehensive information about an individual's search behavior and the ground truth self-esteem. Through our experiments, we found that both of the baseline models achieved higher F1 score (SVM: 0.71 & logistic regression: 0.70) using Lambda Features than using Category Features (SVM: 0.59 & logistic regression: 0.50). One potential explanation for such behavior is that Lambda Features capture both the semantic and temporal component, unlike the Category Features. In terms of information content, Category Features are "subsets" of Lambda Features. More importantly, models when trained on SE-augmented Lambda Features always performed the best (SVM: 0.75& logistic regression: 0.74), see Fig. 6 Table (c). This verifies our hypothesis that there is a interplay between the SE and the history of SI of an individual. The 95% CI was estimated by creating $1,000$ bootstrap subsamples.

For the graphical model, we learn the posterior distribution of parameters via MCMC sampling. We sampled 1000 sets of parameters from the learned posterior distribution. For each set of the sampled parameters, we evaluated the model performance on the test set and achieved an average (over all 1k sampled parameters) F1 score of 0.83. The detailed precision and recall for each class, the macro/weighted average F1 scores, and the ROC AUC are reported in Fig. 7.

To summarize, no matter what model is used, features constructed with SE always outperformed those without SE, supporting the hypothesis that SE can be one of the many factors associated with the history of suicidal thinking.

(a)

Test set N = 24	PSI	NPSI	Macro Avg.	Weighted Avg
Precision	0.76	1	0.88	0.87
Recall	1	0.64	0.81	0.83
F1 score	0.87	0.78	0.82	0.83

ROC AUC score: **0.81**

(b)

Test set N = 24	True PSI	True NPSI
Pred PSI	7	0
Pred NPSI	4	13

Confusion Matrix

Fig. 7. The performance of the graphical model classifiers on the PSI detection task.

5 Discussion

Suicide is a worldwide public health concern of growing importance. Extensive research has been done in an effort to predict suicide ideation, attempts, and death and to provide appropriate interventions. Although predicting suicide ideation does not by itself solve the problem of suicide attempt prediction, it is an important first step. Furthermore, our proposed probabilistic model provides a more expansive parsing of the conceptual framework of self-esteem, suicide ideation, and online behavior that has been employed in prior work. To the best of our knowledge, we are the first to demonstrate that self-esteem, along with online search logs, can detect history of suicide ideation. While this paper is about detecting history of SI, eventually what we really care about is either a) detecting current SI or b) predicting SI in the near future.Those questions remain part of our future work.

Limitations: One shortcoming of this study is the limited sample size. While our findings are consistent with what might be expected based on traditional work on suicide ideation, attempt, death, and the relationship with self-esteem, they can not be considered definitive. However, our results clearly demonstrate *an observable dependency of history of suicide ideation with daily online activities and self-esteem.*

Secondly, our work focuses on identifying individuals with prior suicide ideation. This is different than predicting "future" self-harm or suicide ideation, which may appear more appealing and useful than detecting individuals with PSI condition. We acknowledge this distinction of our work. However, prior research showed that suicide ideation is one of the key risk factors for suicide death [34]. It is uncertain when or whether someone with PSI condition will have future suicide ideation, however identifying individuals with PSI condition can be a potential first step to flag such population at risk. We acknowledge the importance of predicting future suicide ideation and to do this it requires data collected from an individual from multiple time points. This is one of our future works as we are about to launch a longitudinal study in our community for this task specifically.

Furthermore, the mental health questionnaire for the ground truth self-esteem and history of suicide ideation is collected via self reported survey. Although these instruments are clinically proven state-of-the-art tools, one can not guarantee that a participant may (i) conceal information or choose the more socially acceptable answer rather than being honest; (ii) not be able to access themselves accurately; (iii) individuals from different socio-economic background may have varying interpretation of mental health related phenomena; (iv) have

varying subjective interpretation of the questions; (v) need to recall something from the past which may result in inaccurate response.

Finally, another limitation of this work is that we are strictly using data from Google search engine logs via the Google Takeout interface. One of the prerequisites to participate in our study required participants to have an active Google account and have been using Google search engine for at least 12 months. Hence, someone using a different search engine will not qualify to participate in our study. With regards to searching in incognito mode, it is not clear to us to what extent Google logs queries made on their search engine. In some cases they report that search engine logs may still be visible in incognito mode[4].

Addressing Bias: To limit recruitment bias, our study advertisements and consent materials used very generic wording, along the lines of " *help us learn about mental health using online data*," without specific mention of SE or SI. We acknowledge that it is impossible to completely eliminate the fact that an individual may conceal information while answering suicidal ideation questions in a survey or never queries search engines about topics that may raise an alarm. Hence, it may be nearly impossible to get data without noises and uncertainty.

Privacy & Ethical Considerations: Given the sensitive nature of both suicide and research using search logs, there are unique ethical dimensions related to works in this area, especially around the issues of privacy and personal safety. The study protocol for this work has been approved by the Institutional Review Board of our institution, in order to ensure that sufficiently rigorous human subjects protections are in place to address these privacy and safety concerns. We also acknowledge that ethical challenges will arise if and when applications based on our methods are deployed.

There are significant ethical challenges at the micro (individual), meso (family and local community), and macro (regional, state, and national) levels posed by this line of work. A detailed discussion of these myriad concerns is beyond the scope of the current paper. We focus instead on the main ethical challenges posted by this work at the micro level. In particular, there are potential threats to individual privacy and safety when data such as search logs are used for secondary purposes. Users likely never intended such secondary uses of their personal data, and may often refuse to allow it, if given direct control over data they generate or contribute to.

A further issue is the question of whether and when it is ethical to intervene a person's life on the basis of a suicide-risk signal that is necessarily imperfect. SI is not a perfect predictor of suicide. The ultimate decision to intervene must, we believe, be made by people who understand both the complexity of suicide and the power and limitations of any automated technology. After all, the above two issues do not exhaust all potential ethical concerns. Understanding and developing approaches to the myriad ethical implications of work on inferring mental health from online behavior remains an active and growing area of research [5].

[4] `https://support.google.com/chrome/answer/7440301?co=GENIE.Platform\%3DAndroid&hl=en#`

5.1 Conclusion

In this paper, we show that histories of suicide ideation can be identified from online search logs, along with using ground truth self-esteem. This is an empirical demonstration that supports the hypothesis from clinical psychology that self-esteem is correlated with suicide ideation. We employed both the semantic of the search content category and the temporal recurrence & frequency to construct vector representations which is used to build model for identifying individuals with history of suicide ideation. We look forward to expanding our analysis across different forms of online activities and eagerly explore signals of other forms of mental health related phenomena.

References

1. Adler, N., Cattuto, C., Kalimeri, K., Paolotti, D., Tizzoni, M., Verhulst, S., Yom-Tov, E., Young, A.: How search engine data enhance the understanding of determinants of suicide in india and inform prevention: observational study. Journal of medical Internet research **21**(1), e10179 (2019)
2. Arria, A.M., O'Grady, K.E., Caldeira, K.M., Vincent, K.B., Wilcox, H.C., Wish, E.D.: Suicide ideation among college students: A multivariate analysis. Archives of Suicide Research **13**(3), 230–246 (2009)
3. Ayers, J.W., Althouse, B.M., Allem, J.P., Rosenquist, J.N., Ford, D.E.: Seasonality in seeking mental health information on google. American journal of preventive medicine **44**(5), 520–525 (2013)
4. Bailey, E., Mühlmann, C., Rice, S., Nedeljkovic, M., Alvarez-Jimenez, M., Sander, L., Calear, A.L., Batterham, P.J., Robinson, J.: Ethical issues and practical barriers in internet-based suicide prevention research: a review and investigator survey. BMC medical ethics **21**, 1–16 (2020)
5. Chancellor, S., Birnbaum, M.L., Caine, E.D., Silenzio, V., De Choudhury, M.: A taxonomy of ethical tensions in inferring mental health states from social media. In: Proceedings of the Conference on Fairness, Accountability, and Transparency. pp. 79–88. ACM (2019)
6. Colombo, G.B., Burnap, P., Hodorog, A., Scourfield, J.: Analysing the connectivity and communication of suicidal users on twitter. Computer communications **73**, 291–300 (2016)
7. Coppersmith, G., Ngo, K., Leary, R., Wood, A.: Exploratory analysis of social media prior to a suicide attempt. In: Proceedings of the Third Workshop on Computational Linguistics and Clinical Psychology. pp. 106–117 (2016)
8. Creemers, D.H., Scholte, R.H., Engels, R.C., Prinstein, M.J., Wiers, R.W.: Implicit and explicit self-esteem as concurrent predictors of suicidal ideation, depressive symptoms, and loneliness. Journal of behavior therapy and experimental psychiatry **43**(1), 638–646 (2012)
9. De Man, A., Gutierrez, B.: The relationship between level of self-esteem and suicidal ideation with stability of self-esteem as moderator. Canadian Journal of Behavioural Science/Revue canadienne des sciences du comportement **34**(4), 235 (2002)
10. De Ruiter, N.M., Van Geert, P.L., Kunnen, E.S.: Explaining the "how" of self-esteem development: the self-organizing self-esteem model. Review of General Psychology **21**(1), 49–68 (2017)

11. DuPaul, G.J., Weyandt, L.L., O'Dell, S.M., Varejao, M.: College students with adhd: Current status and future directions. Journal of attention disorders **13**(3), 234–250 (2009)
12. of Education (SAVE), S.A.V.: SUICIDE FACTS (2020 (accessed Aug 1, 2020)), `https://save.org/about-suicide/suicide-facts/`
13. Farok, N.H.M., Mahmud, N.: The influence of social media on suicidal ideation: a systematic literature review. Journal of Research in Psychology **2**(1), 4–9 (2020)
14. GlobalStats: Search engine market share worldwide. `https://bit.ly/382vgWD` (2019), accessed: 09-04-2019
15. Google: Google Takeout (2020 (accessed Aug 1, 2020)), `https://takeout.google.com/`
16. Guan, L., Hao, B., Cheng, Q., Yip, P.S., Zhu, T.: Identifying chinese microblog users with high suicide probability using internet-based profile and linguistic features: classification model. JMIR mental health **2**(2), e17 (2015)
17. Gunn III, J.F., Lester, D.: Using google searches on the internet to monitor suicidal behavior. Journal of affective disorders **148**(2), 411–412 (2013)
18. Ji, S., Pan, S., Li, X., Cambria, E., Long, G., Huang, Z.: Suicidal ideation detection: A review of machine learning methods and applications. IEEE Transactions on Computational Social Systems (2020)
19. Jimenez, A., Santed-Germán, M.A., Ramos, V.: Google searches and suicide rates in spain, 2004-2013: correlation study. JMIR public health and surveillance **6**(2), e10919 (2020)
20. Kann, L., McManus, T., Harris, W.A., Shanklin, S.L., Flint, K.H., Hawkins, J., Queen, B., Lowry, R., Olsen, E.O., Chyen, D., et al.: Youth risk behavior surveillance—united states, 2015. Morbidity and Mortality Weekly Report: Surveillance Summaries **65**(6), 1–174 (2016)
21. Kernis, M.H.: Substitute needs and the distinction between fragile and secure high self-esteem. Psychological Inquiry **11**(4), 298–300 (2000)
22. Kroenke, K., Spitzer, R.L., Williams, J.B.: The phq-9. Journal of general internal medicine **16**(9), 606–613 (2001)
23. Liu, D., Baumeister, R.F.: Social networking online and personality of self-worth: A meta-analysis. Journal of Research in Personality **64**, 79–89 (2016)
24. Lowry, R., Crosby, A.E., Brener, N.D., Kann, L.: Suicidal thoughts and attempts among us high school students: trends and associated health-risk behaviors, 1991–2011. Journal of Adolescent Health **54**(1), 100–108 (2014)
25. McGee, R., Williams, S., Nada-Raja, S.: Low self-esteem and hopelessness in childhood and suicidal ideation in early adulthood. Journal of abnormal child psychology **29**(4), 281–291 (2001)
26. Nock, M.K., Borges, G., Bromet, E.J., Alonso, J., Angermeyer, M., Beautrais, A., Bruffaerts, R., Chiu, W.T., De Girolamo, G., Gluzman, S., et al.: Cross-national prevalence and risk factors for suicidal ideation, plans and attempts. The British Journal of Psychiatry **192**(2), 98–105 (2008)
27. Nock, M.K., Borges, G., Bromet, E.J., Cha, C.B., Kessler, R.C., Lee, S.: Suicide and suicidal behavior. Epidemiologic reviews **30**(1), 133–154 (2008)
28. Nuti, S.V., Wayda, B., Ranasinghe, I., Wang, S., Dreyer, R.P., Chen, S.I., Murugiah, K.: The use of google trends in health care research: a systematic review. PloS one **9**(10), e109583 (2014)
29. Orth, U., Robins, R.W., Roberts, B.W.: Low self-esteem prospectively predicts depression in adolescence and young adulthood. Journal of personality and social psychology **95**(3), 695 (2008)

30. Page, A., Chang, S.S., Gunnell, D.: Surveillance of australian suicidal behaviour using the internet? (2011)
31. Pearl, A., Kiang, A., Bailon, J.: Data loss prevention (dlp) methods by a cloud service including third party integration architectures (Oct 18 2016), uS Patent 9,473,532
32. Pineiro, C.R.: Social media use and self-esteem in undergraduate students. Rowan University (2016)
33. Purcell, K.: Search and email still top the list of most popular online activities. Pew Internet & American Life Project **9**, 1–15 (2011)
34. Ribeiro, J., Franklin, J., Fox, K.R., Bentley, K., Kleiman, E.M., Chang, B., Nock, M.K.: Self-injurious thoughts and behaviors as risk factors for future suicide ideation, attempts, and death: a meta-analysis of longitudinal studies. Psychological medicine **46**(2), 225–236 (2016)
35. Rosenberg, M.: Rosenberg self-esteem scale (rse). Acceptance and commitment therapy. Measures package **61**, 52 (1965)
36. Rosenberg, M.: Self-concept from middle childhood through adolescence. Psychological perspectives on the self **3** (1986)
37. Salvatier, J., Wiecki, T.V., Fonnesbeck, C.: Probabilistic programming in python using PyMC3. PeerJ Computer Science **2**, e55 (apr 2016). https://doi.org/10.7717/peerj-cs.55, `https://doi.org/10.7717/peerj-cs.55`
38. Schreiber, F., Bohn, C., Aderka, I.M., Stangier, U., Steil, R.: Discrepancies between implicit and explicit self-esteem among adolescents with social anxiety disorder. Journal of Behavior Therapy and Experimental Psychiatry **43**(4), 1074–1081 (2012)
39. Sowislo, J.F., Orth, U.: Does low self-esteem predict depression and anxiety? a meta-analysis of longitudinal studies. Psychological bulletin **139**(1), 213 (2013)
40. Sueki, H.: Does the volume of internet searches using suicide-related search terms influence the suicide death rate: Data from 2004 to 2009 in japan. Psychiatry and clinical neurosciences **65**(4), 392–394 (2011)
41. van Tuijl, L.A., de Jong, P.J., Sportel, B.E., de Hullu, E., Nauta, M.H.: Implicit and explicit self-esteem and their reciprocal relationship with symptoms of depression and social anxiety: A longitudinal study in adolescents. Journal of behavior therapy and experimental psychiatry **45**(1), 113–121 (2014)
42. Turecki, G., Brent, D.A.: Suicide and suicidal behaviour. The Lancet **387**(10024), 1227–1239 (2016)
43. Vanhalst, J., Luyckx, K., Scholte, R.H., Engels, R.C., Goossens, L.: Low self-esteem as a risk factor for loneliness in adolescence: Perceived-but not actual-social acceptance as an underlying mechanism. Journal of abnormal child psychology **41**(7), 1067–1081 (2013)
44. WHO: Preventing suicide: A global imperative. World Health Organization (2014)
45. Wild, L.G., Flisher, A.J., Lombard, C.: Suicidal ideation and attempts in adolescents: associations with depression and six domains of self-esteem. Journal of adolescence **27**(6), 611–624 (2004)
46. Yang, A.C., Tsai, S.J., Huang, N.E., Peng, C.K.: Association of internet search trends with suicide death in taipei city, taiwan, 2004–2009. Journal of affective disorders **132**(1), 179–184 (2011)
47. Youn, S.J., Trinh, N.H., Shyu, I., Chang, T., Fava, M., Kvedar, J., Yeung, A.: Using online social media, facebook, in screening for major depressive disorder among college students. International Journal of Clinical and Health Psychology **13**(1), 74–80 (2013)

48. Zaman, A., Acharyya, R., Kautz, H., Silenzio, V.: Detecting low self-esteem in youths from web search data. In: The World Wide Web Conference. pp. 2270–2280. ACM (2019)